Examiner only

7. Jupiter has 79 moons in orbit around it. The table shows data for 4 of its moons.

Name of moon	Mean diameter (km)	Mean temperature (°C)	Orbit radius (km)	Orbit time (days)
Io	3 660	–163	421 700	1.8
Europa	3 120	–171	671 000	3.6
Ganymede	5 260	–163	1 070 400	7.2
Callisto	4 820	–139	1 882 700	16.7

Use information from the table above to answer the following questions.

(a) (i) State which moon has the highest mean temperature. .... [1]

(ii) For the solar system, as the orbit radius increases around the Sun, the mean temperature of the planets generally decreases.
Explain why the orbit radius of the moons around Jupiter does not affect their temperatures in the same way. [2]

....

....

....

(iii) Peter states that as Callisto has the longest orbit time it must be the largest moon.
Determine whether Peter's claim is correct. [1]

....

....

(iv) Peter correctly notices that Ganymede has an orbit time that is exactly double Europa's.
Peter suggests that as the orbit time **doubles**, the orbit radius also **doubles**.
Use **only** the data for Ganymede and Europa to determine whether Peter's claim is true. [1]
Space for calculations.

....

....

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(b) (i) Two of the statements listed below are correct. One correct statement has already been ticked.
Tick (✓) **one more box** to show the other correct statement. [1]

An Astronomical Unit (AU) is the mean distance that separates the Earth and the Sun.	✓
A light year is a measurement of time.	
A light minute is the distance travelled by light in 60 seconds.	
A light year is smaller than a light second.	

(ii) Both Earth and Jupiter travel in elliptical paths around the Sun.
As they orbit the Sun the closest distance between Jupiter and Earth is 588 000 000 km. This is equivalent to 3.92 AU.

Calculate the distance, in km, that separates the **Earth and the Sun**. [2]
(1 AU = Earth to Sun distance)

Distance = ... km

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END OF PAPER